

facturers of this new technology will do sufficient volumes to drastically reduce prices, and faster DDR3 memory modules appear—which will increase the (current) nominal performance advantage that DDR3 has over DDR2. Intel's upcoming Nehalem and their X58 chipsets will be DDR3 only. AMD's Sandtiger, the successor to their Phenom processor range, may also go the DDR3 way, but that largely depends on Intel—the market leader. Intel made DDR2 the success it was, and it seems they're going to lay it to rest soon as well.

The other newcomer we're spotlighting is an exciting new VRAM standard—GDDR5, and the successor for the ageing GDDR3, which has been around since GeForce 6800 Ultra / Radeon X800XT days. But wait! Where's GDDR4? Defunct, largely, thanks to the industry having taken a jump from GDDR3 to GDDR5. GDDR4 with its high latencies and comparable-to-GDDR3-speeds is largely a no-go, and other than the odd acceptance by AMD/ATi on their X1950XTX and HD3870, has been largely left alone. Both NVIDIA and ATI have tried all the tricks to up memory bandwidth, which include bumping up clock speeds and/or increasing the bus width. GDDR3 has hit the MHz threshold at 2.4 GHz, and developing wider-than-256-bit memory is very costly. Enter GDDR5—a new standard that vendors like Qimonda (formerly Infineon) and Samsung claim can easily hit speeds of 5 GHz, while consuming less power—no more than 1.5v at whatever speed. Add to that extra features geared towards reliability and you've got a sure shot recipe for a successor.

DDR3: In Pursuit Of 8 Lane Expressways

DDR3 is being offered in the same 240 pin configuration that DDR2 is available with, but the two are incompatible and the physical notches on the modules' PCBs have different positions to avoid mistakenly slotting of the modules in incorrect slots. The first thing that's radically different about DDR3 is the improved clock speeds—to the tune of 2000 MHz whereas the fastest DDR2 modules could do no more than 1200 MHz. Its early days and the speeds of DDR3 could go as high as 2400 MHz—an exciting prospect. Unfortunately latency (wasted clock cycles) also increases with higher clocks, and DDR3 has much higher latencies than DDR2 which somewhat nullifies the clock advantage. The increase in clock frequency is possible because DDR3 has double the prefetch rate than DDR2. The DDR3 core is capable of prefetching 8 bits in a clock, while DDR2 could only manage 4 bits per clock. Doubling the prefetch is possible by doubling the number of internal banks between the DRAM core and the I/O buffer. This speeds up things and further reduces latency due to paral-

DDR2 VS DDR3

	DDR2 SDRAM	DDR3 SDRAM
Data transfer rate	400-800 Mbit/s	800-1600 Mbit/s
Voltage	1.8 V $\pm$ 0.1 V	1.5 V $\pm$ 0.075 V
Banks	4/8	8
Prefetch	4 bits	8 bits
Burst Length	BL=4,8	BL=4,8
Topology	Conventional T	Fly-by
CAS Latency	3-5	6-10
Chips packaging	60 BGA for x4/x8	78 BGA for x4/x8
	84 BGA for x16	96 BGA for x16
Chips capacity	256 Mbit-4 Gbit	512 Mbit-8 Gbit

lelism. While the advantages of DDR3 may not be apparent at 1066 MHz (the current maximum official frequency for DDR2), at speeds of 1333 MHz and above, the newcomer should start baring its teeth.

From the table above we see that DDR3 has an improvement in clock speeds, data throughput and power consumption, which (clock to clock) should reduce proportionately due to lower operating voltages. The BGA packaging remains unchanged although DDR3 chips have a higher pin count. These extra pins are mainly for improving signal integrity at faster clock speeds. Its highly undesirable, (a gross understatement), for memory modules to have corrupt data in their buffers as a result of signal loss—which itself occurs once clock speeds are bumped up.

The architecture adopted by DDR3 is called *Fly-by* and it features additional goodies like on-module signal termination, and other flow stabilisation commands—to aid data integrity. Unlike DDR2's architecture where signals are sent simultaneously, DDR3 follows a more serial approach by sending signals sequentially—one by one to the memory chips. Therefore the internal working of DDR3 memory in terms of data reading and writing is totally different from DDR2. This sequential operation needs some sort of counter or timer in order to link past, current and future operations that are based on the same process. The solution—integrate such a sequencing device on to the memory controller itself. This gives the controller the ability to manage capsules of data and match them to their relevant processes. The technical term for this is called *Data Levelling*. The signal protocols for DDR3 have changed too—as a direct result of the much higher frequencies. These obvious differences mean that DDR3 RAM also has an entirely new set of signal protocols to cope with the higher memory bus frequency.

64-bit operating systems mean greater memory addressing capacities than before. This combined with more bandwidth intensive applications dictates that memory density should increase, and not just its frequency. With DDR2